Volume of boxes and prisms

Day 19 • 2026-12-01 • 12:45 PM–2:00 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ & REASON

Which formula finds a rectangular prism’s volume?

A. $V = 2(L + W)$

B. $V = L \times W$

C. $V = L + W + H$

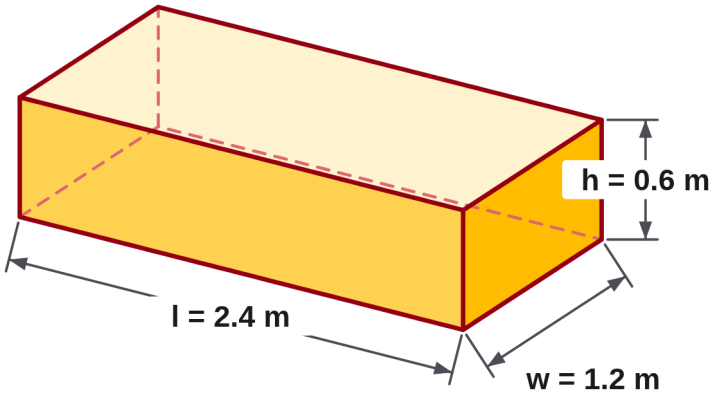
D. $V = L \times W \times H$

Choice (A–D) _____ **Working / reason** _____

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

02

Find the volume of the Shape Canvas prism measuring 2.4 m by 1.2 m by 0.6 m.



Shape Canvas

Answer (m³) _____ **Working / reason** _____

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

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03 Use Numeracy Hub to check a box-volume calculation. Which unit belongs to a volume calculated from dimensions in metres?

A. L/m

B. m^3

C. m^2

D. m



Numeracy Hub

Choice (A–D)

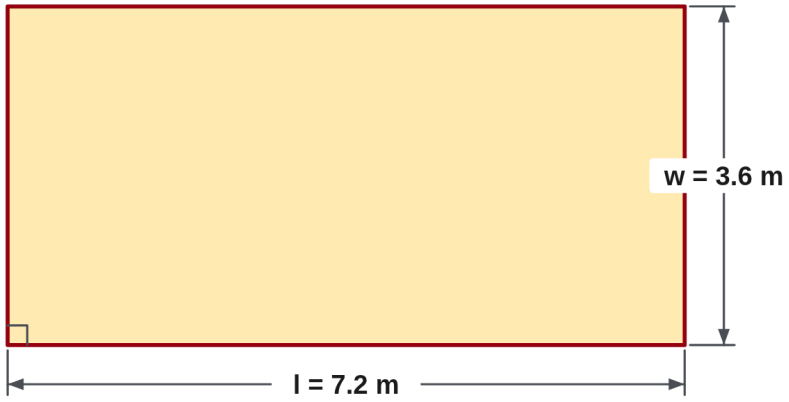
Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Volume of boxes and prisms

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04 The Shape Canvas floor is 7.2 m by 3.6 m. A slab over it is 0.1 m thick. Find its volume.



Shape Canvas

Answer (m³)

Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

05 How many litres are in 1 m³?

A. 10 L

B. 100 L

C. 1000000 L

D. 1000 L

Choice (A–D)

Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

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06 CALCULATE

A box is 3 m long, 2 m wide and 0.75 m high. Find its volume.

Answer (m³)	Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

07 A container’s internal dimensions are 600 mm by 400 mm by 300 mm. Find its capacity in litres.

Answer (L)	Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

08 A triangular prism has cross-sectional base 1.2 m, perpendicular height 0.8 m, and length 3 m. Find its volume.

Answer (m³)	Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

09 A slab is 5 m by 3 m and 120 mm thick. Find its volume in m³.

Answer (m³)	Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

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10 Convert 2.75 m³ to litres.

Answer (L)

Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

11 APPLY & CHECK

A learner calculates the capacity of a thick-walled tank using outside dimensions. What should be used instead?

A. Only the outside height

B. The outside perimeter

C. The wall thickness alone

D. Internal dimensions

Choice (A–D)

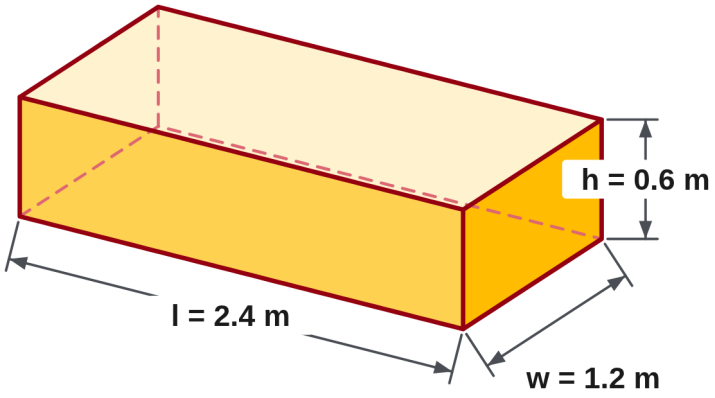
Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Volume of boxes and prisms

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12 Use the Shape Canvas prism’s 1.728 m³ volume. How many litres is that?



Shape Canvas

Answer (L)

Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

13 A slab calculation gives 3.6 m³. Add 5% order allowance before delivery-size rounding.

Answer (m³)

Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

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14 A prism’s length doubles while its cross-section stays unchanged. What happens to volume?

A. It doubles

B. It quadruples

C. It stays unchanged

D. It halves

Choice (A–D)

Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

15 Which check detects an answer accidentally labelled m^2 for a box volume?

A. Round the numerical answer

B. Change the answer to perimeter

C. Ignore the label if the calculator agrees

D. Count the three multiplied length units

Choice (A–D)

Working / reason

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172